

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Importance of biodiversity: resources & ecosystem services - Set 1

Mark scheme

AQA spec 3.1.2.1; The living environment; textbook pp. 10-19

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Copying the structural / functional adaptations of living organisms in the design of manufactured items (1) • Aircraft wingtip fins copied from splayed wingtip feathers of soaring birds, reducing turbulence and fuel use (1) • Humpback whale flipper tubercles copied on ship rudders and wind turbine blades (tighter turns, rotation at lower wind speeds) (1) • Shark skin ridges reducing friction / bacterial adhesion (surface coatings, hospital surfaces) (1) • Termite mound convection copied for natural ventilation of buildings; hollow strutted bird bones in lightweight roofs; gecko toe adhesion; burr hooks - Velcro; lotus leaf self-cleaning glass; spider silk for airbags / body armour; bat / dolphin echolocation - ultrasound scanners (1)	4
01.2	<i>1 mark for each point.</i> • Plants produce toxic chemicals (eg alkaloids) to defend themselves against herbivores; in controlled doses these affect human physiology (1) • Examples: morphine / codeine from poppies; quinine from cinchona bark (malaria); aspirin from willow bark; taxol from yew (cancers); diosgenin from Mexican yam (contraceptives, cortisone) (1) • Animal examples accepted: AZT from a marine sponge (HIV), shark liver lipids (tumours) (1)	3
01.3	<i>1 mark for each point.</i> • Only a very small proportion of species have been studied for medicinal substances so unknown species may contain valuable chemicals (1) • Conserving the habitat protects all its species, including those not yet investigated (and their interdependent relationships) (1)	2

Question	Marking guidance	Marks
01.4	<i>1 mark.</i> - Squid giant nerve cells - nerve function / heart disease, stroke; - marsupials (kangaroo, wallaby pouch) - embryo development; hippo skin secretion - sunscreen / burns; sea urchin embryos - teratogen testing; armadillo - leprosy; marine sponge proteins - transplant rejection (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-15; AQA spec 3.1.2.1.1-3.1.2.1.2.	
02.1	<i>1 mark for each point.</i> - Domesticated crops were bred from a small number of original plants so they have a small gene pool / low genetic diversity (1) - Selective breeding for yield, appearance or shelf life may have lost other useful genes (eg taste, disease resistance) (1) - Genetically similar plants share the same susceptibilities so a new disease or environmental change can affect the whole crop (1)	3
02.2	<i>1 mark for each point.</i> - A region with a high concentration of the wild relatives of important crops - usually where the crop was first domesticated (eg wheat / barley in the Middle East; potato, tomato in South America; rice in India; maize in Central America) (1) - They hold the largest gene pools / most genetic variety for future crop breeding (disease resistance, drought and salt tolerance, yield) (1) - Many are in regions where environmental degradation / habitat loss threatens the wild plants (1)	3
02.3	<i>1 mark for each point.</i> - A large population may all descend from a few closely related individuals so they are genetically similar (1) - Similar adaptations mean all are susceptible to the same change (disease, climate) (1) - Greater risk of inbreeding so disadvantageous recessive genes are expressed (eg albinism, poor muscle development in lion cubs); so species should be protected across their whole range to retain regional genes (1)	3
02.4	<i>1 mark.</i> Morona bean (drought resistant, soya-like protein); yuca tree (drought resistant nuts, poor soils); potato bean (high protein); perennial maize / Kernza (no annual replanting, less ploughing / erosion); salt-tolerant spinach relatives (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 15-17; AQA spec 3.1.2.1.2.	

Question	Marking guidance	Marks
03.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Photosynthesis and aerobic respiration balance O₂ and CO₂ in a dynamic equilibrium; evapotranspiration adds water vapour forming clouds and rain; microbes drive the nitrogen and other biogeochemical cycles so wastes do not accumulate and nutrients are recycled (2) • Detritivores, fungi and bacteria decompose dead organic matter releasing nutrients and forming humus that binds soil; organic acids weather rock forming new soil; roots bind soil, reducing erosion and regulating water flow / flooding (2) • Species provide food, pollination (bees, moths, birds, bats - more reliable than wind, especially in forests), seed dispersal (animals carry larger seeds to suitable sites, faeces as fertiliser, gut passage stimulates germination) and habitat provision (trees as nest sites and canopy microclimate; mollusc shells for hermit crabs) (2)	6
03.2	<i>1 mark for each point.</i> • Its nectar is at the end of a 30 cm tube reached only by the long tongue of the sphinx moth, which carries pollen only between orchids of the same species (1) • Advantage: no pollen wasted on other species and the moth has an exclusive food source; risk: if either species declines the other cannot survive (1)	2
03.3	<i>1 mark for each point.</i> • They are not ruminants so seeds are not destroyed by repeated chewing / pass through intact; they eat the fruit of most tree species (1) • Not territorial so they carry seeds over large distances / areas (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 18-19; AQA spec 3.1.2.1.3.	

Question	Marking guidance	Marks
04.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Resources: wood, fibres (cotton, wool, silk), oils, fuels (wood, biofuels), food; new food species (morama bean, perennial cereals, eland, bison) better adapted to local conditions • Biomimetics: vehicle design, architecture, structures, adhesion, materials, ultrasound • Medicines from plants and animals (examples); most species unstudied - argument for habitat conservation • Physiological research using wildlife species; wildlife as pest control agents (ladybirds, <i>Encarsia</i>, <i>Cactoblastis</i>, Bt) • Genetic resources: crop wild relatives, Vavilov centres, gene pools • Ecosystem services: atmospheric composition, hydrological and biogeochemical cycles, soil maintenance, pollination, seed dispersal, habitat provision • Ethical / aesthetic / recreational arguments; the unknown value of species not yet investigated (precautionary principle) • Evaluation: which arguments are most persuasive to decision makers; economic valuation vs intrinsic value; judgement	9
	Model answer - Level 3 (8 marks)	
	Human society depends on other species first as resources: wood, fibres such as cotton, wool and silk, oils, fuels and, above all, food, with wild species such as the morama bean, perennial cereals, eland and bison offering new crops and livestock better adapted to local conditions. Biomimetics copies adaptations into vehicle design, architecture, structures, adhesives, materials and ultrasound; medicines such as taxol, quinine and AZT came from plants and animals, and physiological research on marsupials, hippos and sponges continues to yield medical insights. Wild species are pest-control agents - ladybirds, <i>Encarsia</i>, <i>Cactoblastis</i>, <i>Bacillus thuringiensis</i> - and crop wild relatives in Vavilov centres hold the genes future crops will need.	
	Less visibly, other species provide the services that keep the planet habitable: regulating atmospheric composition, driving the hydrological and biogeochemical cycles, maintaining soil, pollinating crops, dispersing seeds and providing habitat. Most species have never been investigated, so their value is unknown, and there are ethical, aesthetic and recreational reasons for conservation that owe nothing to usefulness.	

Question	Marking guidance	Marks
	Which arguments persuade decision-makers? Economic valuation of resources and services speaks the language of policy and has been most effective, but it undervalues the services that go unnoticed because they operate in dynamic equilibrium and it can never capture the value of species not yet studied. The strongest case combines the precautionary logic of unknown value with the intrinsic argument: conserving biodiversity protects both what we know we need and what we cannot yet know.	
	Weaker answer - Level 1 (3 marks)	
	Biodiversity is important for humans in many ways. We get food, wood and cotton from plants and animals. Many medicines come from plants. Bees pollinate our crops so we need to conserve them. Animals and plants are also nice to look at and people like to visit nature. We should conserve biodiversity because we might need species in the future.	
	What makes the difference	
	• Level 1 names categories without examples; Level 3 gives specific, accurate examples in every category.	
	• Level 3 includes ecosystem services, genetic resources and pest control that Level 1 omits.	
	• 'Discuss' at the top level means weighing the arguments against each other - Level 3 evaluates which arguments persuade and why; Level 1 does not.	
04.2	<i>1 mark.</i> • The variety and abundance of life (species, genes, habitats) in an area / on Earth (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	
05.1	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Potato bean (N America): high protein content (1) • Morama bean (southern Africa): protein content similar to soya, drought resistant (1) • Yeheb tree (Somalia): edible nuts, drought resistant, grows on poor soils (1) • Perennial maize (Mexico) / Kernza: no need to buy seed each year, less ploughing so less soil erosion (1) • American bison: graze grassland too poor for cattle (1) • Common eland: browse and graze so can be kept with cattle without severe competition (1)	2
05.2	<i>1 mark for each point.</i> • Indigenous species are better adapted to local conditions such as climate, pests / diseases and soil (1) • So they have higher survival rates / productivity is not reduced, as it is when introduced species are kept in areas to which they are not adapted (1)	2
05.3	<i>1 mark.</i> • They grow in soils with a high salt content, so their genes could be used in breeding programmes to increase the salt tolerance of crops grown on soils salinised by irrigation (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11; AQA spec 3.1.2.1.1.	

Question	Marking guidance	Marks
06.1	<i>1 mark for each point.</i> • Ship rudders that turn in tighter circles / aerogenerator blades that start turning at lower wind speeds (1) • Water-repelling surface of lotus flowers (1) • Coatings for aircraft and ships that reduce fuel consumption / hospital coatings that bacteria cannot stick to (1) • Hooked burrs on seeds that stick to animal fur (1)	4
06.2	<i>1 mark for each point.</i> • The mound absorbs sunlight and warms; hot air inside rises, creating a convection current that draws stale air out of the lower parts (solar-driven ventilation) (1) • Air blowing over the mound has a lower pressure which also draws air out (1) • Copied in shopping complexes / office blocks to give natural ventilation and cooling without air conditioning, so less energy / electricity is used (1)	3
06.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Marsupials: young develop in the pouch so are easier to study than a human foetus, helping understanding of developmental problems (1) • Hippopotamus skin secretes hipposudoric acid (natural sunscreen and antimicrobial) which may improve treatment of burns (1) • Marine sponges produce proteins that prevent rejection of grafts, being developed to prevent rejection of transplanted organs (1)	2
06.4	<i>1 mark.</i> • They are light but strong: hollow with internal struts that prevent bending / breaking, copied in lightweight bridges and roofs (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11-14; AQA spec 3.1.2.1.1.	
07.1	<i>1 mark for each point.</i> • The processes (eg regulation of atmospheric gases by photosynthesis and respiration) act in a dynamic equilibrium and cancel each other out (1) • Because there is no overall change their actions go unnoticed / their importance is not appreciated until they are lost (1)	2
07.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Invertebrates, fungi and bacteria break down / decompose dead organic matter, releasing nutrients for plant growth (1) • The organic matter and humus produced hold the soil together (reducing erosion) and regulate the water cycle, giving more even river flow and less flooding (1) • Organic acids from decomposition aid the weathering of rock, producing more soil and releasing more nutrients (1) • Microbes carry out the processes of the carbon, nitrogen and phosphorus cycles so wastes do not build up and nutrients are not depleted (1)	3
07.3	<i>1 mark for each point.</i> • Trees reduce wind velocity so wind pollination would be ineffective / unreliable over long distances (1) • Insects search over long distances for flowers so dispersed populations can be pollinated, and less pollen needs to be produced (energy saving); specialised flowers reduce pollen being wasted on other species (1)	2

Question	Marking guidance	Marks
07.4	<i>1 mark for each point.</i> • Dispersal is not haphazard: animals live in the same habitat as the plant so are likely to carry seeds to locations where they can survive (1) • Seeds can be carried longer distances than by wind and can be larger, containing more nutrients for the seedling (1) • Seeds passing through the gut are deposited in faecal material which acts as a fertiliser, and passage through the intestine may stimulate germination (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 18-19; AQA spec 3.1.2.1.3.	
08.1	<i>1 mark for each point.</i> • Plants produce chemicals (often alkaloids) that are toxic to the herbivores that would eat them (1) • In carefully controlled amounts these chemicals have beneficial physiological effects in humans (1)	2
08.2	<i>1 mark for each point.</i> • Only a very small proportion of species have been studied for medicinal substances, so it is not known which species will be useful (1) • Conserving the habitat protects all the species in it, including those not yet studied and the species they depend on (1)	2
08.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Marsupials (kangaroo / wallaby): young develop in the pouch so are easier to study than a human foetus - developmental problems (1) • Hippopotamus: hipposudoric acid skin secretion (sunscreen / antimicrobial) - burns treatment (1) • Marine sponges: proteins preventing graft rejection - organ transplants (1)	2

Question	Marking guidance	Marks
08.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Resources: wood, fibres, oils, fuels, food; new food species adapted to local conditions (morama bean, yeheb, perennial maize, Kernza, bison, eland) • Biomimetics: whale flipper tubercles, shark skin, termite-mound ventilation, bird bones, gecko adhesion, burrs / Velcro, lotus self-cleaning, spider silk • Medicines and physiological research; genetic resources - crop wild relatives, Vavilov centres, gene pools, inbreeding • Ecosystem services: atmospheric composition, hydrological cycle, biogeochemical cycles, soil maintenance; interspecies relationships - food, pollination, seed dispersal, habitat provision • Evaluation: services go unnoticed because of dynamic equilibrium; unknown future value; economic vs moral / intrinsic arguments; conservation also needed for aesthetic and cultural reasons • Justified conclusion	9
	Model answer - Level 3 (9 marks)	
	Other species supply resources we use directly: wood, fibres such as cotton and wool, oils, fuels and food. Because most farmed species were domesticated long ago and are often kept where they are poorly adapted, wild species such as the drought-resistant morama bean, the yeheb tree that grows on poor soils, perennial maize that needs no ploughing, and bison that graze grassland too poor for cattle offer new, better-adapted food sources. Wild species also carry genetic resources: crop wild relatives in Vavilov centres hold genes for disease resistance and salt tolerance that inbred commercial varieties lack.	
	Biomimetics copies adaptations into technology - humpback flipper tubercles in turbine blades, shark skin in friction-reducing coatings, termite-mound convection in naturally ventilated buildings, hollow bird bones in lightweight roofs. Medicines such as taxol, quinine, AZT and diosgenin came from wild species, and physiological research on marsupials, hippo skin and sponge proteins is helping with foetal development, burns and transplant rejection. Since only a tiny fraction of species has been studied, the unknown future value argues for conserving whole habitats.	

Question	Marking guidance	Marks
	Beyond resources, other species provide services we could not replace: regulation of atmospheric gases, evapotranspiration driving rainfall, the carbon, nitrogen and phosphorus cycles run by microbes, soil formation by decomposers, and pollination, seed dispersal and habitat provision. These are unnoticed precisely because they are in dynamic equilibrium, so their economic value is underestimated. On balance the resource and service arguments alone justify conservation, but they are strongest when combined with the intrinsic and aesthetic case, because many species with no known use still sustain the systems we depend on.	
	Weaker answer - Level 1 (3 marks)	
	Biodiversity is important because we get lots of things from other species. We get wood, cotton and food from plants and animals. Some medicines come from plants, like aspirin from willow trees. Animals can help design things, like Velcro was copied from seeds. Other species also do things for us like bees pollinating crops. If we lose species we might lose things that could be useful in the future. So we should conserve biodiversity.	
	What makes the difference	
	Both answers cover resources, medicines, biomimetics and services, but Level 3 develops each with specific, accurate detail (which species, what property, why it matters).	
	The question asks 'to what extent'. Level 3 weighs resource arguments against intrinsic value and reaches a reasoned conclusion; Level 1 asserts 'so we should conserve'.	
	Level 3 explains <i>why</i> services are undervalued (dynamic equilibrium) - an explanation, not just a list.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	